COGKNOWEDGE SOLUTIONS

PostgreSQL
Zero to Production

A hands-on course in 8 sections: 16 practical labs on one shared university database, plus a capstone in a new domain. Concepts come first in the Introduction; every lab after that is practical.

8	16+1	~15.5 h	18
sections	labs + capstone	of lab time	PostgreSQL version

Course Overview: sections, labs and descriptions

Course at a glance

PostgreSQL is a free, open-source object-relational database with more than 35 years of active development, known for reliability, standards-compliant SQL and unusual extensibility. This course teaches it through 16 practical labs and a capstone that share one university domain (students, courses, instructors, enrollments), so every lab builds on data you already know. The capstone moves to a new domain, a multi-tenant marketplace, to prove the skills transfer.

Who it is for

- Developers, analysts and data engineers
- Complete beginners, no database experience needed
- Engineers moving from MySQL, SQLite or MongoDB

Prerequisites

- Basic computer skills
- Comfort typing commands in a terminal
- Python is used only in Lab 7

By the end you can

- Model data and write confident SQL, from SELECT to window functions
- Read query plans and make slow queries fast
- Secure, back up and replicate a server

What each lab includes

- Written walkthrough and runnable hands-on
- Assignment with answer key
- Same kit as the MongoDB Labs course

Difficulty key

BEGINNER	INTERMEDIATE	INTER. TO ADVANCED	ADVANCED	CAPSTONE

The learning path

0 Introduction	1 Foundations	2 Modeling and Analytics	3 Rich Data and Data Movement
4 Performance and Concurrency	5 Programmability and Cost Control	6 Security and Operations	7 Search
8 Capstone			

Circle colour shows the hardest level in each section. Labs are designed to be taken in order; the Requires column lists what each lab builds on. Time estimates are planning figures for a learner working through the walkthrough and hands-on.

Built on PostgreSQL 18

Examples target the current major version (18, September 2025). Features that are new in 18 (asynchronous I/O, skip scan, uuidv7(), virtual generated columns, OAuth 2.0 authentication, OLD and NEW in RETURNING) are called out in the lab; everything else works on recent earlier versions.

Table of Contents

PostgreSQL: Zero to Production | Introduction + 16 labs + capstone | 8 sections | about 15.5 hours of lab time

0	**Introduction**			Concepts and setup
0.1	Table of Contents	OVERVIEW		
0.2	What Is PostgreSQL and How Does It Work?	CONCEPTS		
0.3	Setup Guide	TO BE ADDED		
1	**Foundations**			Tables, SQL basics and the full data lifecycle
1	Create Tables and Query Data	BEGINNER	~40 min	None
2	Update Data and Enforce Constraints	BEGINNER	~40 min	Lab 1
2	**Modeling and Analytics**			Schema design, joins and analytical SQL
3	Model Relationships and Join Tables	INTERMEDIATE	~50 min	Labs 1-2
4	Summarize Data with Aggregates and Subqueries	INTERMEDIATE	~45 min	Labs 1-3
5	Analyze Data with CTEs and Window Functions	INTERMEDIATE	~50 min	Labs 1-4
3	**Rich Data and Data Movement**			Flexible data types and getting data in and out
6	JSONB, Arrays, Ranges and Generated Columns	INTERMEDIATE	~45 min	Labs 1-4
7	Load, Migrate and Move Data	INTERMEDIATE	~50 min	Labs 1-3
4	**Performance and Concurrency**			Fast queries and correct behavior under load
8	Speed Up Queries with Indexes and EXPLAIN	INTER. TO ADVANCED	~55 min	Lab 6
9	Tune Workloads with Stats and Partitioning	INTER. TO ADVANCED	~55 min	Lab 8
10	Transactions, MVCC and Locking	ADVANCED	~45 min	Labs 1-4
5	**Programmability and Cost Control**			Logic in the database and spending less to run it
11	Functions, Triggers and Extensions	ADVANCED	~50 min	Labs 1-5
12	Optimize Cost: Storage, Bloat and Right-Sizing	INTER. TO ADVANCED	~60 min	Labs 8-10
6	**Security and Operations**			Protect, recover, observe and scale
13	Secure Postgres with Roles, RLS and TLS	ADVANCED	~50 min	Labs 1-5
14	Back Up, Recover and Replicate Postgres	ADVANCED	~60 min	Labs 1-10
15	Monitor, Tune and Scale Postgres	ADVANCED	~55 min	Labs 8-14
7	**Search**			Ranked, typo-tolerant keyword search inside Postgres
16	Full-Text and Fuzzy Search	INTERMEDIATE	~55 min	Labs 6 and 8
8	**Capstone**			A new-domain, end-to-end project
17	Capstone: Multi-Tenant Marketplace Database	CAPSTONE	~120 min	Labs 1-16

Columns: lab number, title, level, estimated time, prerequisite labs.

0 Introduction

Start here | concepts and setup

The Introduction holds all the concept material, the same way the RAG Labs and MongoDB Labs READMEs explain what the technology is before any lab starts. Labs stay practical and assume you have read it.

0.1

Table of Contents

OVERVIEW

The course map: every section and lab with its level, time estimate and prerequisites, plus a short description of what each lab covers.

0.2

What Is PostgreSQL and How Does It Work?

CONCEPTS

A single concept guide covering everything a learner should understand before touching SQL. It is organized in seven parts:

- 0.2.1 **What Is PostgreSQL?.** An open-source object-relational database that began as the POSTGRES project at UC Berkeley in 1986 and has been ACID-compliant since 2001. Includes a one-line analogy, the PostgreSQL License and why teams choose it.
- 0.2.2 **SQL and the Relational Model.** Tables, rows, columns and keys explained from scratch, and why relating data across tables beats copying it.
- 0.2.3 **Core Building Blocks.** The hierarchy every lab uses: cluster, database, schema, table, row. Also roles and the system catalogs.
- 0.2.4 **How PostgreSQL Works.** What happens between typing a query and seeing a result: client and server, the postmaster, backend processes, shared buffers, the write-ahead log (WAL), MVCC and background workers such as autovacuum.
- 0.2.5 **PostgreSQL vs Other Databases.** Side-by-side comparison with MySQL, SQLite and MongoDB, and when each is the better fit.
- 0.2.6 **Key Features.** Data types (JSONB, arrays, ranges, UUID), integrity constraints, indexing, replication, security, extensibility and full-text search, plus what is new in PostgreSQL 18.
- 0.2.7 **Real-World Use Cases and Course Map.** Where Postgres runs in production, and how the 16 labs, the capstone and the 8 sections fit together: labs on one running university database, then a capstone in a new domain.

0.3 Setup Guide

TO BE ADDED

This lesson is intentionally left empty for now. The content will be written in a later step.

1 Foundations

Section 1 | 2 labs | Tables, SQL basics and the full data lifecycle

1 Create Tables and Query Data

BEGINNER
~40 min
Requires: None

Connect with psql, build the university database and its first tables, load sample rows and answer questions with SELECT.

Covers: psql and pgAdmin basics; databases and schemas; data types; CREATE TABLE; INSERT; SELECT, DISTINCT, WHERE, ORDER BY, LIMIT; LIKE, IN, BETWEEN; NULL handling; CASE; everyday functions

2 Update Data and Enforce Constraints

BEGINNER
~40 min
Requires: Lab 1

Finish the data lifecycle: change and remove rows safely, load data without duplicates, and make the database reject bad data by itself.

Covers: UPDATE, DELETE, TRUNCATE; RETURNING; upserts with ON CONFLICT and MERGE; ALTER TABLE; NOT NULL, UNIQUE, CHECK; primary keys, identity columns and uuidv7()

2 Modeling and Analytics

Section 2 | 3 labs | Schema design, joins and analytical SQL

3 Model Relationships and Join Tables

INTERMEDIATE
~50 min
Requires: Labs 1-2

Design the student, course and enrollment model with foreign keys, learn what makes a schema good or bad, then combine the tables with every kind of join.

Covers: Foreign keys and ON DELETE; one-to-many and many-to-many; ER diagrams; normalization (1NF to 3NF); denormalization and anti-patterns; INNER, LEFT, RIGHT, FULL, CROSS and SELF joins; UNION, INTERSECT, EXCEPT

4 Summarize Data with Aggregates and Subqueries

INTERMEDIATE
~45 min
Requires: Labs 1-3

Turn rows into answers: totals and averages per group, questions nested inside questions, subtotals in one query and reports that fill gaps in dates.

Covers: COUNT, SUM, AVG, MIN, MAX; GROUP BY, HAVING, FILTER; array_agg, string_agg, percentiles; subqueries, EXISTS, ANY / ALL, LATERAL; GROUPING SETS, ROLLUP, CUBE; dates, time zones, intervals, generate_series

5 Analyze Data with CTEs and Window Functions

INTERMEDIATE
~50 min
Requires: Labs 1-4

Break big questions into readable steps, rank and compare rows without collapsing them, and save results as views for reuse.

Covers: CTEs and recursive queries; window functions (ranking, running totals, LAG / LEAD, frames); month-over-month reports; views, materialized views, REFRESH ... CONCURRENTLY

3 Rich Data and Data Movement

Section 3 | 2 labs | Flexible data types and getting data in and out

6 JSONB, Arrays, Ranges and Generated Columns

INTERMEDIATE

~45 min

Requires: Labs 1-4

Store flexible course metadata as JSON, model lists, ranges and fixed choices, and let the database compute values for you.

Covers: JSONB operators, jsonpath, JSON_TABLE; GIN indexes on JSONB; arrays, ranges and multiranges, enums; composite types; stored and virtual generated columns (virtual by default in 18); UUID and other specialty types

7 Load, Migrate and Move Data

INTERMEDIATE

~50 min

Requires: Labs 1-3

Get real data into Postgres fast and safely, move it between databases, and query outside sources as if they were tables.

Covers: COPY and \copy; the bulk-load recipe (one transaction, indexes after, ANALYZE); pg_dump and parallel pg_restore; foreign data wrappers (postgres_fdw, file_fdw); migrating from MySQL, SQLite or MongoDB; versioned schema migrations; loading from Python with psycopg 3

4 Performance and Concurrency

Section 4 | 3 labs | Fast queries and correct behavior under load

8 Speed Up Queries with Indexes and EXPLAIN

INTER. TO ADVANCED

~55 min

Requires: Lab 6

Find slow queries, read their plans, add the right index and prove the improvement with numbers.

Covers: B-tree, multicolumn, covering, partial and expression indexes; GIN, GiST, SP-GiST, BRIN; skip scan (18); EXPLAIN (ANALYZE, BUFFERS); scan and join types; index-only scans; what indexes cost on writes

9 Tune Workloads with Stats and Partitioning

INTER. TO ADVANCED

~55 min

Requires: Lab 8

Go from one slow query to a slow system: find the worst offenders, help the planner choose better and split huge tables so they stay fast.

Covers: pg_stat_statements; planner statistics, ANALYZE, extended statistics; keyset pagination; fixing N+1 patterns; range, list and hash partitioning with pruning; parallel query, JIT and async I/O (18)

10 Transactions, MVCC and Locking

ADVANCED

~45 min

Requires: Labs 1-4

Reproduce a seat-booking race condition with two sessions, then fix it. See why readers never block writers and how locks and isolation levels decide who wins.

Covers: ACID; BEGIN, COMMIT, ROLLBACK, savepoints; MVCC and snapshots; isolation levels and retries; row locks, SELECT FOR UPDATE, SKIP LOCKED; advisory locks; deadlocks

5 Programmability and Cost Control

Section 5 | 2 labs | Logic in the database and spending less to run it

11 Functions, Triggers and Extensions

ADVANCED
~50 min
Requires: Labs 1-5

Move repeatable logic next to the data: write functions and procedures, keep an automatic audit trail, push events to applications and add extensions.

Covers: SQL and PL/pgSQL functions; procedures; triggers and event triggers; LISTEN / NOTIFY; extensions (pgcrypto, pg_cron, PostGIS overview)

12 Optimize Cost: Storage, Bloat and Right-Sizing

INTER. TO ADVANCED
~60 min
Requires: Labs 8-10

Find where the money goes and cut it without hurting performance: reclaim wasted space, drop unused indexes, expire old data cheaply and size the server to the real workload.

Covers: table and index size analysis; bloat, VACUUM and autovacuum tuning; unused and duplicate indexes (pg_stat_user_indexes); efficient data types and column order; retention with partition drop and archiving; TOAST and compression; right-sizing CPU, memory and connections from metrics; PgBouncer pooling; managed-cloud cost levers (instance class, storage type and IOPS, read replicas vs standbys, reserved capacity, serverless trade-offs)

6 Security and Operations

Section 6 | 3 labs | Protect, recover, observe and scale

13 Secure Postgres with Roles, RLS and TLS

ADVANCED
~50 min
Requires: Labs 1-5

Give each user exactly the access they need, enforce per-tenant row access and lock down how clients connect.

Covers: roles, GRANT / REVOKE, default privileges; row- and column-level security; pg_hba.conf; SCRAM-SHA-256 (MD5 deprecated); TLS and certificates; OAuth 2.0 (18); SQL injection and prepared statements; pgcrypto; audit logging

14 Back Up, Recover and Replicate Postgres

ADVANCED
~60 min
Requires: Labs 1-10

Make backups you have actually tested, rewind a database to a chosen moment, and keep a live copy on a standby.

Covers: WAL and checkpoints; pg_dump, pg_restore, pg_basebackup; WAL archiving and PITR; restore testing, RPO and RTO; streaming replication, hot standby, promotion; logical replication; replication slots

15 Monitor, Tune and Scale Postgres

ADVANCED
~55 min
Requires: Labs 8-14

Know what the database is doing right now, tune the settings that matter and plan for growth and upgrades.

Covers: pg_stat views and pg_stat_activity; logs and alert-worthy metrics; postgresql.conf tuning (shared_buffers, work_mem, effective_cache_size); read scaling and sharding overview; minor and major upgrades with pg_upgrade; running Postgres in Docker and on managed cloud services

7 Search

Section 7 | 1 lab | Ranked, typo-tolerant keyword search inside Postgres

16

Full-Text and Fuzzy Search

Build a course search that ranks results, highlights matches, forgives typos and understands language, with no external search engine.

Covers: tsvector, tsquery, websearch_to_tsquery; stemming and stop words; weighted ranking (setweight, ts_rank); highlighted snippets (ts_headline); GIN indexes for search; fuzzy matching and autocomplete with pg_trgm; unaccent; ICU collations (case- and accent-insensitive); combining search with SQL filters

INTERMEDIATE

~55 min

Requires: Labs 6 and 8

8 Capstone

Section 8 | 1 lab | A new-domain, end-to-end project

17

Capstone: Multi-Tenant Marketplace Database

Leave the university behind and build the database for a multi-seller online marketplace: sellers see only their own data, stock never oversells, orders stay fast as they grow, products are found with ranked, typo-tolerant search, and the whole system is backed up, monitored and cost-reviewed.

Covers: schema and constraints (Labs 1-3); seed data with COPY (7); revenue analytics with CTEs and windows (4-5); product attributes in JSONB (6); ranked, typo-tolerant product search (16); indexes, EXPLAIN and monthly partitioned orders (8-9); oversell-proof checkout and a SKIP LOCKED fulfilment queue (10); audit triggers (11); retention and cost review (12); per-seller row-level security and roles (13); backup, PITR and a replica (14); health checks (15)

CAPSTONE

~120 min

Requires: Labs 1-16

Sources and how this outline was built

The first outline had 56 small lessons. They were merged into practical labs, the concept topics moved into the Introduction, and the course was expanded with labs on data movement, cost optimization, security and operations. The structure follows the RAG Labs and MongoDB Labs courses.

About PostgreSQL
https://www.postgresql.org/about/
Feature list, history, licence and community facts for the Introduction.

PostgreSQL 18 Documentation: Table of Contents
https://www.postgresql.org/docs/18/static/
Official Parts I to VIII; used to order topics from tutorial to administration.

PostgreSQL 18 Released
https://www.postgresql.org/about/news/postgresql-18-released-3142/
Source for the version 18 highlights (async I/O, skip scan, uuidv7, virtual generated columns, OAuth, OLD/NEW in RETURNING).

PostgreSQL 18 Docs: Populating a Database
https://www.postgresql.org/docs/18/populate.html
Basis for the bulk-load recipe in Lab 7 (COPY, indexes after load, maintenance_work_mem, parallel restore, ANALYZE).

PostgreSQL 18 Docs: Routine Vacuuming
https://www.postgresql.org/docs/18/routine-vacuuming.html
Basis for bloat, autovacuum and wraparound content in Lab 12.

Percona: Reducing PostgreSQL Costs in the Cloud
https://www.percona.com/blog/reducing-postgresql-costs-in-the-cloud/
Cost levers for Lab 12: right-sizing, storage choices, read replicas vs standbys, pooling, vacuum tuning, serverless trade-offs.

Neon: 6 Tips to Optimize Storage Costs for Postgres
https://neon.com/blog/6-tips-to-optimize-storage-costs-for-your-postgres-databases
Unused indexes, efficient data types, column ordering, retention and autovacuum for Lab 12.

Citus Data: Use Cases
https://www.citusdata.com/use-cases
Capstone rationale: multi-tenant SaaS, real-time analytics, time series and high-throughput transactional workloads.

STACKIT: PostgreSQL Core Features and Use Cases
https://docs.stackit.cloud/products/databases/postgresql-flex/basics/core-features-and-use-cases-of-postgresql-flex/
Capstone rationale: financial systems, BI, geospatial, multi-tenant SaaS (row-level security), time series, ERP and content search.

Dataquest: SQL Projects
https://www.dataquest.io/blog/sql-projects/
Survey of common SQL portfolio projects, used to avoid a generic capstone.

W3Schools PostgreSQL Syllabus
https://www.w3schools.com/POSTGRESQL/postgresql_syllabus.php
Beginner coverage benchmark: tables, CRUD, query syntax, joins, grouping.

Udemy: PostgreSQL Bootcamp, Complete Beginner to Advanced
https://www.udemy.com/course/postgresql-bootcamp-complete-beginner-to-advanced-course/
37-lecture SQL-focused syllabus.

Udemy: Master PostgreSQL Database from Beginner to Advanced
https://www.udemy.com/course/master-postgresql-database-from-beginner-to-advanced/
18-section syllabus; only the first 10 sections are publicly visible.

Educative: The Art of PostgreSQL
https://www.educative.io/courses/the-art-of-postgresql/course-overview
42-module course; informed data types, modeling, concurrency and extensions coverage.

RAG Labs and MongoDB Labs (local folders)
Structural templates: concepts in the Introduction/README, numbered practical labs with level, time and prerequisites, one running domain, capstone last.

Notes: Lab titles are 47 characters or fewer. Times, prerequisites and the university and marketplace domains are proposals. Walkthroughs, hands-on files, assignments and the Setup Guide are not written yet.